

Question Paper

Exam Date & Time: 20-Nov-2019 (10:00 AM - 01:00 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

I Semester of M. Sc.(Biochemistry/Microbiology/Biotechnology) Examination, November,2019

Course code & Name MS 712 & Biochemistry

Date: 20/11/2019 Day: Wednesday

BIOCHEMISTRY [MS712]

Marks: 70

Duration: 180 mins.

MCQ

Answer all the questions.

Section Duration: 30 mins

Choose the most correct answer for the following questions (1-20).

- 1) Some proteins contain additional amino acids that arise by modification of an amino acid already present in a peptide, examples include (1)
[4 -hydroxy proline.](#) [5- hydroxy Lysine.](#) [Gamma Amino Butyric Acid.](#) [All of the above.](#)
- 2) Which of following is an anomeric pair? (1)
[D-Glucose and L- Glucose.](#) [α-D-Glucose and β-D- Glucose.](#) [D-Glucose and D- Fructose.](#) [α-D-Glucose and β -D- Fructose.](#)
- 3) In the absence of oxygen, the primary purpose of fermentation is to: (1)
[Produce amino acids for protein synthesis.](#) [Generate a proton gradient for ATP synthesis.](#) [Oxidize glucose to generate reduce electron carriers.](#) [Regenerate NAD+ from NADH allowing glycolysis to continue.](#)
- 4) Which of the following enzymes is present in the snake venom? (1)
[Phospholipase A1.](#) [Phospholipase A2.](#) [Phospholipase B.](#) [Phospholipase D.](#)
- 5) A 9-year old boy presents with altered mental status, heart failure, and muscle weakness. His serum level of ketones and glucose are abnormally low. He is diagnosed with primary carnitine deficiency. In which of the following is carnitine directly involved? (1)
[Transport of fatty acyl CoA.](#) [β-Oxidation.](#) [Cytosol.](#) [Activation of fatty acids.](#)
- 6) The terminal electron acceptor during mitochondrial respiration: (1)
[H₂O.](#) [NAD⁺.](#) [ATP.](#) [O₂.](#)
- 7) Chylomicrons are synthesized in (1)
[Blood.](#) [Liver.](#) [Pancreas.](#) [Intestine.](#)
- 8) A noncompetitive inhibitor of an enzyme-catalyzed reaction (1)
[increases K_M and increases V_{max}.](#) [increases K_M and reduces V_{max}.](#) [reduces K_M and increases V_{max}.](#) [reduces K_M and reduces V_{max}.](#)
- 9) A 20-year girl has been fasting for religious reason for several days. Her brain has reduced its need for glucose by using which of the following substances as an alternate source of energy? (1)
[Fatty acids.](#) [Glycerol.](#) [Beta hydroxyl butyrate.](#) [Beta Carotene.](#)
- 10) Which of the followings is a common nitrogen acceptor for all reactions involving transaminases? (1)
[α ketoglutarate.](#) [Pyruvate.](#) [Oxaloacetate.](#) [Acetoacetate.](#)
- 11) In a 56-year-old man, who has been diagnosed with cirrhosis of liver, ammonia is not getting detoxified and damage brain. Which of the following amino acids can covalently bind ammonia, transport and store in a nontoxic form? (1)
[Aspartate.](#) [Glutamate.](#) [Serine.](#) [Cysteine.](#)
- 12) Which of the following is not a type of lipid? (1)
[Fat.](#) [Cholesterol.](#) [Chitin.](#) [Prostaglandins.](#)
- 13) There are four enzymes of gluconeogenesis that circumvent the irreversible steps in glycolysis. (1)

When starting with the substrate pyruvate or lactate they are

Pyruvate carboxylase, phosphoenolpyruvate carboxykinase, fructose-1,6-bisphosphatase, and glucose-6-phosphatase.	Hexokinase, phosphofructokinase-1, phosphofructokinase-2 and pyruvate kinase.	Glycerol kinase, glycerol-3-phosphate dehydrogenase, fructose-2,6-bisphosphatase, and glucose-6-phosphatase.	Amino transferase, phosphoenolpyruvate carboxykinase, fructose-2,6-bisphosphatase, and glucose-6-phosphatase.
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- 14) A reversible inhibitor that can bind to either E alone or the ES complex is referred to as a (1)
[Competitive inhibitor.](#) [Uncompetitive inhibitor.](#) [Suicide inhibitor.](#) [Non-competitive inhibitor.](#)
- 15) The activities of many enzymes, membrane transporters and other proteins can be quickly activated or inactivated by phosphorylation of specific amino acid residues. This regulation is called (1)
[Allosteric inhibition.](#) [Covalent modification.](#) [Feedback inhibition.](#) [Competitive inhibition.](#)
- 16) ω -oxidation of fatty acids takes place at (1)
[Mitochondria.](#) [Peroxisome.](#) [Endoplasmic Reticulum.](#) [Peroxisome and Mitochondria.](#)
- 17) Which membrane fatty acids are precursors for eicosanoids? (1)
[C20 and C22 saturated fatty acids.](#) [C18 and C22 polyunsaturated fatty acids.](#) [C20 and C22 monounsaturated fatty acids.](#) [C20 and C22 polyunsaturated fatty acids.](#)
- 18) A 22-year old male type-1 diabetic faints after injecting himself with insulin. He is administered Glucagon and rapidly recovers consciousness. Glucagon induces activity of: (1)
[Glycogen synthase.](#) [Glycogen phosphorylase.](#) [Glucokinase.](#) [Hexokinase.](#)
- 19) What is the name of the drug which inhibits Na^+/K^+ pump across the cell membrane? (1)
[Taxol.](#) [Vinblastine.](#) [Ouabain.](#) [Quinone.](#)
- 20) Which of the following kinetic parameters best describes how well suited a specific compound functions as a substrate for a particular enzyme? (1)
[K_m.](#) [V_{max}.](#) [K_{cat}.](#) [K_{cat}/K_m.](#)

SECTION-I

Answer 5 out of 8 questions.

Answer the following questions as directed (Any 5).

- What is meant by inhibition of an enzyme? Draw individual Lineweaver-Burk plots to differentiate different type of reversible inhibition. (4)
- What are the differences between transamination and oxidative deamination? Discuss briefly. (4)
- What is the importance of the primary structure of a protein? What is the difference between alpha-helix and beta-sheet protein conformations? Are there any changes in the primary structure when a protein is denatured? (4)
- How do you sequence a protein? Discuss briefly. (4)
- What is feeder pathway for Glycolysis? Discuss briefly. (4)
- Give names of some Uncommon Amino Acids and mention also their important functions. (4)
- Write a short note on Suicide inhibition. (4)
- Compare and contrast "fat" (triglycerides) and fatty acids. Draw the condensed structural formula of any monounsaturated fatty acid. (4)

SECTION-II

Answer 10 out of 14 questions.

Answer the following questions as directed (Any 10).

- Calculate the net number of ATPs produced when one 14-carbon fatty acid salt is activated, enters the mitochondrion, and undergoes complete β oxidation. Include the ATP formed from acetyl-CoA in the citric acid cycle, and ATP from all of the reduced coenzymes that are produced. (3)
- What is the net change in ATP, NADH, and FADH_2 from the passage of two acetyl-CoA through the citric acid cycle? Which two compounds donate their electrons to electron transport chain? Which molecule is the final electron acceptor of the electron transport chain? (3)

- 3) What would the result be if both glycolysis and gluconeogenesis pathways were operating simultaneously and at the same rate? What is the general term that defines how the situation described above is avoided ? (3)
- 4) List the attractive interactions involved in secondary, tertiary, and quaternary protein structure. (3)
- 5) What are anaplerotic reactions? Discuss briefly. (3)
- 6) Do changes in the primary, secondary or tertiary structure of a protein normally have consequences for their function? Do all proteins have a quaternary structure? (3)
- 7) A new protein of unknown structure has been purified. Gel filtration chromatography reveals that the native protein has a molecular weight of 300,000 daltons (300 kDa). Chromatography in the presence of 6M guanidine hydrochloride yields only a peak for a protein of 75 kDa. Chromatography in the presence of 6M guanidine hydrochloride and 10mM β -mercaptoethanol yields peaks for proteins of 40 kDa and 35 kDa. Explain what can be determined about the structure of this protein from these data and why. (3)
- 8) List three "strategies" used by enzymes to catalyze chemical reactions: (3)
- 9) In sickle cell anemia, a hereditary disease, one amino acid is substituted by another in one of the four polypeptide chains of hemoglobin. In this case are all of the structural levels of the protein modified? (3)
- 10) What compound acts as an inhibitor of glycolysis? Is the conversion of pyruvate to lactate an oxidation or a reduction? When this reaction takes place in a cell, what is the oxidizing agent? (3)
- 11) Write down the possible catabolic fates of the pyruvate formed in Glycolysis. (3)
- 12) Write as short note on amphibolic nature of TCA cycle. (3)
- 13) How Eicosanoids act as signaling molecule? Describe briefly with examples. (3)
- 14) What are glycosides? Give examples and functions of two glycosides. (3)

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